



Asymptotic expansion for a function containing irrational exponents

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I'm trying to find an asymptotic expansion of a function containing irrational exponents in the form of a power series (which also might contain irrational exponents). It works correctly if I request only the leading term:



```
Asymptotic[(1 + x + x^√2)^√2, x -> ∞]
(* x^2 *)
```



But if I request more terms, unfortunately, it just returns the original function rather than a power series expansion:

```
Asymptotic[(1 + x + x^√2)^√2, x -> ∞, SeriesTermGoal -> 7]
(* (1 + x + x^√2)^√2 *)
```

So far, I had to write this function myself as a workaround for this problem:

```
asymptotic[f_, n_] :=
  f - Nest[s |->
    s - (x^# Limit[s/x^#, x -> ∞] & )@Simplify@Limit[x D[s, x]/s, x -> ∞], f, n];

asymptotic[(1 + x + x^√2)^√2, 7]
(* x^2 + (13 - 9 √2)/12 x^(6 - 4 √2) +
  (-1 + 2 √2/3) x^(5 - 3 √2) + (2 - √2) x^(3 - 2 √2) +
  (1 - 1/√2) x^(4 - 2 √2) + √2 x^(2 - √2) + √2 x^(3 - √2) *)
```

It's obviously very limited because it cannot handle functions with more complicated asymptotics (such as a power series plus/times an additional exponential or logarithmic term).

Is there a better/simpler way to do this? Why can't the built-in [Asymptotic](#) handle these cases?

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edited Oct 16, 2022 at 1:20

asked Oct 15, 2022 at 23:55



Vladimir Reshetnikov
6,651 32 83

- 1 ▲ I like your `asymptotic` function. But as you point out, it does not work for say `(1+x+Log[x])^Sqrt[2]`, and a general function would have to automatically detect which method is appropriate. In your example, two other methods come to mind. First, [binomial series](#). Second, note that $f(x) = (1 + x + x^a)^b$ satisfies the differential equation $f'(x) = g(x)f(x)$ where $g(x) = \frac{b(1+ax^{a-1})}{1+x+x^a}$, one can write down an expansion for g using geometric series, then use that to build up an expansion for f , but this also requires work. – [user293787](#) Oct 16, 2022 at 4:25

Because I have a trinomial, I will have to apply the formula for the binomial series twice, and I will end up with a double sum, correct? And the terms in the double sum will no longer be linearly ordered by their growth rate, and I will have to find which term of the double sum I need to pull next when more terms in the asymptotic expansion are requested. Is my understanding correct here?

– [Vladimir Reshetnikov](#) Oct 17, 2022 at 3:35

- 1 ▲ Yes, a double sum. If $a < 1$ then x is the largest of $1, x, x^a$ for large x and therefore $(1 + x + x^a)^b = x^b(1 + x^{-1} + x^{-(1-a)})^b = \sum_{m,n \geq 0} \binom{b}{m,n,b-m-n} x^{b-m-n(1-a)}$ where the sum is over integers m and n , and $\binom{b}{m,n,b-m-n}$ is a **Multinomial** coefficient (have not checked the details). To order the summands by their exponent $b - m - n(1 - a)$, you could first use `Table` to generate all summands that you want to include, then `SortBy` them. – [user293787](#) Oct 17, 2022 at 3:56 ✎

- 1 ▲ Sorry, in your case $a > 1$ then it would be $(1 + x + x^a)^b = x^{ab}(1 + x^{-a} + x^{-(a-1)})^b = \sum_{m,n \geq 0} \binom{b}{m,n,b-m-n} x^{ab-am-n(a-1)}$. For example, for $a = b = \sqrt{2}$ then the summand $m = 0, n = 2$ is `Multinomial[0,2,Sqrt[2]-2] x^(4-2*Sqrt[2])` which you can check matches one of your terms. – [user293787](#) Oct 17, 2022 at 4:05 ✎

I guess `Limit[x D[s, x]/s, x -> ∞]` does something similar internally to identify the exponent of the leading term. – [Vladimir Reshetnikov](#) Oct 17, 2022 at 13:52